Little Lab Coats

☀️ The Sun Warms Everything ☀️

Physical Science — Sunshine & Warmth — Unit 2, Lesson 1 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Physical Science — Sunshine & Warmth 🔢 **Lesson:** 1 of 6

Which will get warmest in the sun? Let's find out!

📋 **NGSS Standard:**
K-PS3-1 — Make observations to determine the effect of sunlight on Earth's surface

🎯 Learning Objective

Students will investigate the Sun Warms Everything, make observations during the lesson activity, and explain what they learned using evidence from the lesson.

🎯 The Big Idea

The sun is a giant ball of energy — and that energy travels all the way to Earth as **light**. When sunlight hits something, it can warm it up! But not everything warms up at the same rate. **Dark colors** absorb (soak up) more sunlight and get warmer faster. **Light colors** reflect (bounce back) more sunlight and stay cooler. Wet things also behave differently than dry things in the sun.

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🌑 Shadow Science 🌑

Physical Science — Sunshine & Warmth — Unit 2, Lesson 2 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Physical Science — Sunshine & Warmth 🔢 **Lesson:** 2 of 6

Light travels in straight lines — when something blocks it, a shadow forms!

📋 **NGSS Standard:**
K-PS3-1 — Make observations to determine the effect of sunlight on Earth's surface

🎯 Learning Objective

Students will investigate shadow Science, make observations during the lesson activity, and explain what they learned using evidence from the lesson.

🎯 The Big Idea

Shadows form when an object blocks sunlight. But shadows aren't just interesting to look at — **shadow length tells us how directly the sun's rays are hitting Earth!** A short shadow means the sun is high in the sky and rays hit the ground directly. Direct rays mean more concentrated warmth. A long shadow means the sun is at a low angle and rays are spread out — less warming per spot.

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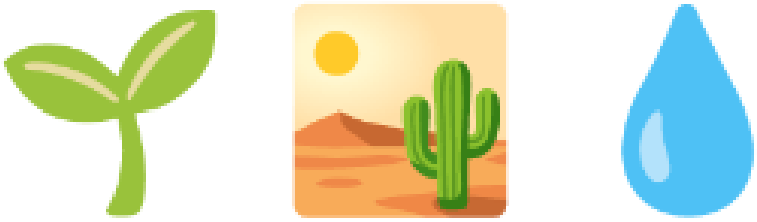
🌍 Warming Up: Dark vs. Light Surfaces 🌍

Physical Science — Sunshine & Warmth — Unit 2, Lesson 3 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Physical Science — Sunshine & Warmth 🔢 **Lesson:** 3 of 6



Soil, sand, and water — three surfaces, three different ways of warming up!

📋 **NGSS Standard:**
K-PS3-1 — Make observations to determine the effect of sunlight on Earth's surface

🎯 Learning Objective

Students will investigate warming Up: Dark vs. Light Surfaces, make observations during the lesson activity, and explain what they learned using evidence from the lesson.

🎯 The Big Idea

Earth has many different surfaces — soil, grass, sand, pavement, water, ice. Each one warms up at a **different rate** when sunlight hits it. Dark surfaces like soil absorb more heat and warm quickly. Light-colored surfaces like sand warm more slowly. Water warms the slowest of all! Understanding this helps us understand weather, climate, and even why cities can feel much hotter than the countryside.

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🧊 The Melting Race! 🧊

Physical Science — Sunshine & Warmth — Unit 2, Lesson 4 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Physical Science — Sunshine & Warmth 🔢 **Lesson:** 4 of 6

Three ice cubes, three conditions — who will melt first?

📋 **NGSS Standard:**
K-PS3-1 — Make observations to determine the effect of sunlight on Earth's surface

🎯 Learning Objective

Students will investigate the Melting Race, make observations during the lesson activity, and explain what they learned using evidence from the lesson.

🎯 The Big Idea

Sunlight can cause solids to melt by warming them. But **where** the ice is placed and **what color** it sits on makes a huge difference in how fast it melts! In this experiment, we put ice cubes in three different conditions and have a melting race. The results connect everything we've learned so far: dark colors absorb more heat, direct sunlight warms more than shade, and reflective surfaces can protect against warming.

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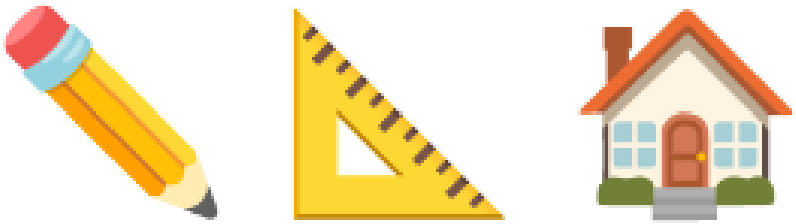
🏗️ Design a Sun Shelter — Part 1: Plan! 🏗️

Physical Science — Sunshine & Warmth — Unit 2, Lesson 5 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Physical Science — Sunshine & Warmth 🔢 **Lesson:** 5 of 6



Every great building starts with a plan. Today we design — next lesson we build!

📋 **NGSS Standard:**
K-PS3-2 — Use tools and materials provided to design and build a structure that will reduce the warming effect of sunlight on an area

🎯 Learning Objective

Students will plan ideas for **"Design a Sun Shelter"** — Part 1: Plan, make observations during the activity, and explain their choices using evidence from the lesson.

🎯 The Big Idea

We've spent four lessons learning HOW sunlight warms surfaces. Now it's time to use that knowledge to **solve a real problem**! Scientists and engineers use what they know to design solutions. Today, students will think like engineers: identify the problem, gather ideas, choose materials, make a plan, and draw their design before they build it. *Planning before building leads to better results!*

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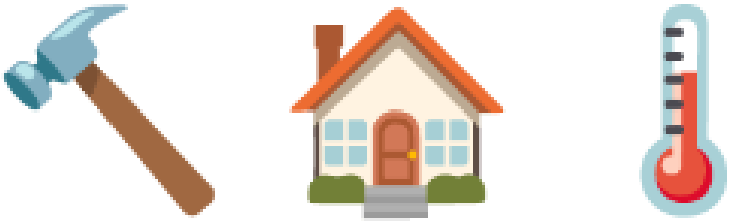
🏗️ Design a Sun Shelter — Part 2: Build & Test! 🏗️

Physical Science — Sunshine & Warmth — Unit 2, Lesson 6 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Physical Science — Sunshine & Warmth 🔢 **Lesson:** 6 of 6 ⭐



Time to build! Does your shelter really keep cool?

📋 **NGSS Standard:**
K-PS3-2 — Use tools and materials provided to design and build a structure that will reduce the warming effect of sunlight on an area

🎯 Learning Objective

Students will build and test ideas for **"Design a Sun Shelter"** — Part 2: Build & Test, make observations during the activity, and explain their choices using evidence from the lesson.

🎯 The Big Idea

Scientists and engineers don't just plan — they **build, test, and improve**! Today, students take their Lesson 5 design plans and bring them to life. After building, they test whether their shelter really does reduce the warming effect of sunlight. Then comes the most important engineering step: **iteration** — looking at what worked and what didn't, and thinking about how to improve the design.

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💡 **Key vocabulary:** build, test, evaluate, iterate, compare, shelter, insulate, reflect, improve, engineering design

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